

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1540

Austin Dam 138 kV
July 20, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1540. The Interconnection Customer proposes a 179.6 MW natural gas combined cycle facility interconnecting at the Austin Dam 138 kV substation in the Pennsylvania area. The request was received on August 03, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 578 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$52,183,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CENTRAL BEACH – FITZPATRICK 138.0 kV Line	TALENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV TX	220.7	318	18.4
PAXON – CENTRAL BEACH 138.0 kV	TALENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV Transformer Bank	219.7	313	18.4
PAXON – S LAKE GENEVA 138.0 kV	TALENERGY MARTINS CRK LLC HARRISBURG 500/345kV Auto	209.2	206	10.8
FITZPATRICK – TALENERGY MARTINS CRK LLC HARRISBURG 138.0 kV	TALENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV Transformer	192.3	382	18.4
HUMMELSTOWN – S LAKE GENEVA 138.0 kV	TALENERGY MARTINS CREEK LLC HARRISBURG 500-345 kV TX	173.7	230	10.8
HAT CREEK 2 – VICTORIA TC 7001-7007-7008 138.0 kV	TALENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV TX	168.8	230	8.2
VICTORIA TC 7001-7007-7008 – BARNES LAKE 138.0 kV L	TALENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV Transformer	163.3	252	8.2
RIVER BEND – HUMMELSTOWN SUBSTATION 138.0 kV line	TALENERGY MARTINS CRK LLC HARRISBURG 500/345kV Transformer	161.9	239	10.8
FT GAGE – AUSTIN DAM 138.0 kV L	CORONET SUBSTATION – WESTWOOD GENERATION LLC 500.0 kV	161.7	207	78.5
GRASS CREEK – BARNES LAKE 138.0 kV	TALENERGY MARTINS CREEK LLC HARRISBURG 500/345kV XFMR	160.6	262	8.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GRASS CREEK SUBSTATION – SAN LORENZO 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV XFMR	158.2	271	8.2
KANSAS RIVER PROJECT – METALCAN 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG SUBSTATION 500-345 kV Auto	156.7	304	5.4
SAN LORENZO SUBSTATION – AJ SWOPE 138.0 kV	TALENENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV Autotransformer	154.8	286	8.2
HAT CREEK 2 – HARGROVE RD D.S 138.0 kV	TALENENERGY MARTINS CREEK LLC HARRISBURG 500/345 kV Transformer Bank	154.5	241	8.2
ZACHARY – COLUMBUS 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG 500-345 kV XFMR	151.7	298	5.4
GARBERVILLE SUBSTATION – RIV BEND 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG SUBSTATION 500/345kV Autotransformer	151.1	248	10.8
AJ SWOPE – TALENENERGY MARTINS CREEK LLC WEST SHORE 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG 500/345 kV Transformer	150.5	306	8.2
COLUMBUS – KANSAS RIV PROJECT 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV XFMR	146.7	314	5.4
PAXON – L L WILKINS 138.0 kV	COONET – WESTWOOD GENERATION LLC SUBSTATION 500.0 kV	144.0	169	6.4
AYER – PAXTON CRK COGENERATION 138.0 kV	CORONET – WESTWOOD GENERATION LLC 500.0 kV	142.6	327	36.4
AYER SUBSTATION – TALENENERGY MARTINS CRK LLC HARRISBURG 138.0 kV Line	WESTWOOD GENERATION LLC – CORONET 500.0 kV	140.5	313	36.4
PAXTON CRK COGENERATION – DUMBARTON TAP 138.0 kV	WESTWOOD GENERATION LLC – CORONET 500.0 kV L	138.7	347	36.4
LAKELAND SOLAR ENERGY LLC – 12TH & IRVING 138.0 kV line	PAXTON CREEK COGENERATION – DUMBARTON TAP 138.0 kV line	135.0	173	18.5
DUMBARTON TAP – MACARTHUR 138.0 kV	CORONET – WESTWOOD GENERATION LLC 500.0 kV	132.6	372	51.5
MACARTHUR – FT GAGE 138.0 kV Line	CORONET – WESTWOOD GENERATION LLC 500.0 kV L	129.5	394	51.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FITZPATRICK – CENTRAL BEACH 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,055,000	66% (\$1,363,000)
PAXON – CENTRAL BEACH 138.0 kV	Rebuild line with higher-capacity conductor	\$3,089,000	66% (\$2,049,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PAXON – S LAKE GENEVA 138.0 kV	Rebuild line with higher-capacity conductor	\$1,886,000	72% (\$1,352,000)
FITZPATRICK – TALENENERGY MARTINS CRK LLC HARRISBURG 138.0 kV	Rebuild line with higher-capacity conductor	\$2,506,000	66% (\$1,662,000)
HUMMELSTOWN – S LAKE GENEVA 138.0 kV	Rebuild line with higher-capacity conductor	\$5,788,000	72% (\$4,151,000)
HAT CREEK 2 – VICTORIA TC 7001-7007-7008 138.0 kV	Rebuild line with higher-capacity conductor	\$2,095,000	100% (\$2,095,000)
BARNES LAKE – VICTORIA TC 7001-7007-7008 138.0 kV	Rebuild line with higher-capacity conductor	\$4,214,000	100% (\$4,214,000)
RIVER BEND – HUMMELSTOWN SUBSTATION 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,389,000	72% (\$1,713,000)
FT GAGE – AUSTIN DAM 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,318,000	100% (\$1,318,000)
GRASS CREEK – BARNES LAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,628,000	100% (\$1,628,000)
GRASS CREEK SUBSTATION – SAN LORENZO 138.0 kV	Rebuild line with higher-capacity conductor	\$3,048,000	100% (\$3,048,000)
KANSAS RIV POJECT – METALCAN 138.0 kV	Rebuild line with higher-capacity conductor	\$2,202,000	51% (\$1,128,000)
SAN LORENZO SUBSTATION – AJ SWOPE 138.0 kV	Rebuild line with higher-capacity conductor	\$4,194,000	100% (\$4,194,000)
HAT CRK 2 – HARGROVE ROAD D.S 138.0 kV	Rebuild line with higher-capacity conductor	\$2,148,000	100% (\$2,148,000)
ZACHARY – COLUMBUS 138.0 kV	Rebuild line with higher-capacity conductor	\$3,396,000	51% (\$1,739,000)
GARBERVILLE SUBSTATION – RIV BEND 138.0 kV	Rebuild line with higher-capacity conductor	\$1,200,000	72% (\$860,000)
TALENENERGY MARTINS CRK LLC W SHORE – AJ SWOPE 138.0 kV	Rebuild line with higher-capacity conductor	\$3,618,000	100% (\$3,618,000)
KANSA RIV PROJECT – COLUMBUS 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,683,000	51% (\$2,399,000)
L L WILKINS – PAXON 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,546,000	100% (\$1,546,000)
AYER – PAXTON CRK COGENERATION 138.0 kV	Rebuild line with higher-capacity conductor	\$2,920,000	100% (\$2,920,000)
AYER – TALENENERGY MARTINS CREEK LLC HARRISBURG 138.0 kV	Rebuild line with higher-capacity conductor	\$520,000	100% (\$520,000)
PAXTON CRK COGENERATION – DUMBARTON TAP 138.0 kV	Rebuild line with higher-capacity conductor	\$1,811,000	100% (\$1,811,000)
LAKELAND SOLAR ENERGY LLC – 12TH & IRVING 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,478,000	100% (\$1,478,000)
DUMBARTON TAP – MACARTHUR 138.0 kV	Rebuild line with higher-capacity conductor	\$2,010,000	100% (\$2,010,000)
MACARTHUR – FT GAGE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,219,000	100% (\$1,219,000)
	Total allocated to project		\$52,183,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.